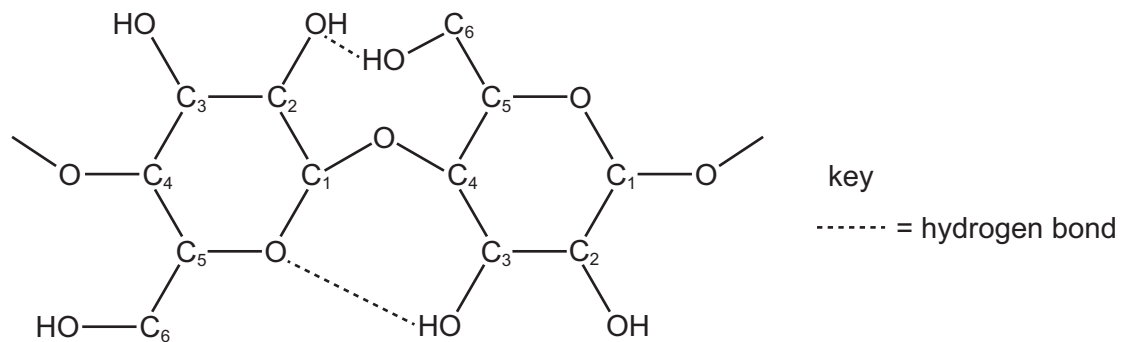


- 9 The diagram shows how the alternating nature of β -glucose monomers along a chain allows hydrogen bonds to form between consecutive monomers.



Hydrogen bonds in cellulose affect the tensile strength (the ability to withstand pulling forces without breaking).

How do the hydrogen bonds shown in the diagram help cellulose function as a suitable material for a cell wall?

- A** The hydrogen bonds add additional tensile strength along individual cellulose molecules within the cell wall.
- B** The hydrogen bonds create stronger crosslinks between adjacent cellulose molecules, adding to the tensile strength of the cell wall.
- C** Stronger hydrogen bonds form between adjacent cellulose molecules, adding to the tensile strength of the cell wall.
- D** Cellulose molecules within a cellulose microfibril have a stronger link between them, increasing the tensile strength of the cellulose microfibrils.